ChatEarthBench

Benchmarking multimodal large language models for Earth observation

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The recent advancements in multimodal large language models (MLLMs) offer new opportunities for Earth observation (EO) tasks by enhancing reasoning and analysis capabilities. However, fair and systematic evaluation of these models remains challenging. Existing assessments often suffer from dataset biases, which can lead to an overestimation of model performance and inconsistent comparisons across MLLMs. To address this issue, we introduce ChatEarthBench, a comprehensive benchmark dataset specifically designed for zero-shot evaluation of MLLMs in EO. ChatEarthBench comprises 10 image-text datasets spanning three data modalities. Importantly, these datasets are unseen by the evaluated MLLMs in our work to enable rigorous and fair zero-shot evaluation across diverse real-world EO tasks. By systematically analyzing MLLM performance across various EO tasks, we provide critical insights into their capabilities and limitations. Our findings offer essential guidance for the development of more robust and generalizable MLLMs for EO applications.

INTRODUCTION

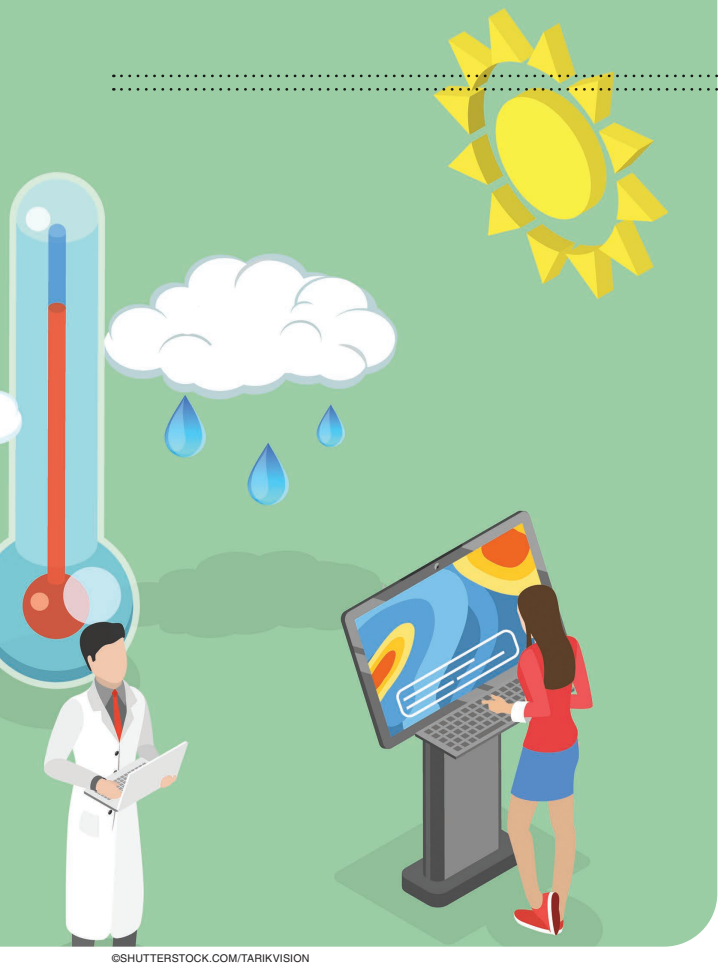
EO data, derived from various geospatial sensors, capture the complex dynamics of Earth's environmental systems and hold vast potential for real-world applications [1], [2], [3]. Despite this potential, challenges exist in utilizing EO data effectively due to the diversity in spatial and temporal scales, data modalities, and inherent complexities of the Earth system [4]. To address these challenges, a range of approaches has been investigated that aims to transform the incomprehensible EO data into meaningful and easily understandable information. In the past few years, natural language processing has been integrated with EO data to explore a variety of vision-language tasks [5], [6] by providing human-like comprehension. For instance, image captioning involves generating textual descriptions of the image content [7]; visual question answering (VQA) can provide answers to questions based on the given image; and visual grounding aims to predict the bounding box for the target object described in a textual input [8].

However, early approaches mainly focus on designing models for specific tasks, meaning that they are task oriented. As a milestone, vision-language models (VLMs) like CLIP [9], RemoteCLIP [10], and SkyScript [11] have been significantly developed, leveraging massive image-text pairs to learn vision-language correlations and enabling zero-shot predictions across diverse tasks [12]. Recently, pretrained LLMs like the GPT series [13], [14], LLaMA series [15], [16], [17], and Qwen series [18], [19] have demonstrated powerful emergent capabilities and reasoning abilities on complex tasks. To take advantage of multiple modalities, MLLMs like LLaVA [20], MiniGPT-4 [21], Qwen-VL [22], InternVL [23], and GPT-4V [14] have become a hot topic in many research domains, including remote sensing. These models usually leverage pretrained vision encoders, such as CLIP ViT [9], to extract visual features. After aligning them with language features, they can be passed to LLMs for various downstream tasks.

Considering the high potential of MLLMs, which provide human-like understanding and reasoning capabilities, they are well suited for a variety of complex EO applications. Consequently, many researchers are actively developing MLLMs tailored for EO tasks that require advanced reasoning capabilities. In recent studies, numerous MLLMs have been proposed to handle diverse EO tasks [5], [24]. These models are usually built by providing instruction-following EO datasets to fine-tune MLLMs. As one of the earliest MLLMs, RSGPT

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[25] focuses on image-level captioning and VQA tasks. However, RSGPT requires task-specific fine-tuning with different datasets for better performance. GeoChat [26] and SkyEyeGPT [27] enhance the capabilities by supporting region-level tasks such as region captioning while also enabling multiturn conversations. More recently, MLLMs for EO are further advanced by adding more tasks at image-level [28], [29], [30], region-level [31], and pixel-level understanding [32] and the temporal level [33], [34] of EO data.

Although there is diversity among the model architectures, fundamental LLMs, and optimization strategies employed in EO MLLMs, these models are mainly based on widely used MLLMs like LLaVA [20]. Compared to architectural innovations, the construction of instruction-following datasets has emerged as the most significant contribution of these EO MLLMs. Consequently, while the model architectures remain similar, the instruction-following datasets differ significantly, making it challenging to directly compare their performance. For example, Table 1 illustrates the diverse data sources employed by different EO MLLM models. The instruction-following datasets are constructed by leveraging existing EO datasets originally designed for specific tasks such as scene classification, object detection, image captioning, and VQA. These datasets are then augmented using LLMs to generate task-specific instruction-text pairs. The table highlights differences in the data sources used to construct the instruction-following datasets. Note that the table lists only the tasks evaluated for MLLMs and does not imply that MLLMs are limited to these tasks.

Regarding the evaluation process, current evaluations rely heavily on existing benchmark datasets and model-specific test sets. While some studies attempt to evaluate multiple models on common datasets, these comparisons are often biased by variations in training data, optimization approaches, and experimental protocols. A more fundamental issue lies in the substantial overlap between training and evaluation datasets across different models. Since evaluation datasets typically incorporate elements from existing EO tasks, data used to evaluate one MLLM may have been used to train another. Figure 1 visualizes the distribution of datasets used for training, evaluation, and their overlap in existing representative EO MLLMs. Although various instruction-following datasets exist, there is a substantial overlap between those used for training and evaluation. In addition, many of these evaluation datasets are not entirely independent but are derived from data sources used in training datasets. For example, SAMRS-Bench [26] is derived from DOTA, DIOR, and FAIR1M, which are also part of some training datasets. This overlap raises potential concerns about the evaluation of the model's generalizability and highlights the need for a more robust evaluation to ensure accurate assessments of EO MLLMs.

This overlap also makes it more challenging to compare existing EO MLLMs using existing datasets. For example, as shown in Table 1, GeoChat [26] has not been fine-tuned on the RSVQA-HR dataset [35], whereas SkyEyeGPT's [27] instruction-following dataset includes RSVQA-HR. As a result, although both models provide test results on RSVQA-HR, it is difficult to determine which model performs better. VRSBench [36] provides a versatile dataset to evaluate multiple vision-language tasks, yet it still contains data (e.g., DOTA and DIOR) that overlap with the training datasets of some existing EO MLLMs, which may make it unfair to use as a benchmark for comparison. This lack of standardization prevents a fair and comprehensive comparison among different MLLMs for EO. Meanwhile, this may also lead to inflated performance claims and limited insights into true model generalization capabilities.

To address these challenges, we present ChatEarthBench, a comprehensive benchmark designed specifically for zero-shot evaluation of EO MLLMs (Figure 2). ChatEarthBench comprises 112K image-text pairs derived from multiple satellite sources, including *Sentinel-1*, *Sentinel-2*, and optical sensors. The benchmark is structured around four key tasks: concise image captioning (CIC), detailed image captioning (DIC), VQA, and multilabel classification (MLC). With 10 distinct subsets spanning multiple spatial resolutions [120], [121], ChatEarthBench deliberately excludes data sources used in training existing EO MLLMs to ensure unbiased evaluation. Through careful curation and task design, ChatEarthBench enables a more standardized assessment of existing EO MLLMs. Furthermore, through comprehensive evaluation and analysis using ChatEarthBench, we identify key challenges and opportunities for advancing the development of EO-focused MLLMs. To summarize, our main contributions are as follows:

- We comprehensively analyze datasets across EO MLLMs, revealing critical gaps in current evaluation practices and

TABLE 1. A COMPREHENSIVE REVIEW OF EXISTING MLLMs FOR EO ON THE INSTRUCTION-FOLLOWING AND EVALUATION DATASETS.

MODEL	INSTRUCTION-FOLLOWING DATASET	INSTRUCTION-FOLLOWING DATA SOURCES	EVALUATION TASKS	EVALUATION DATASETS
RSGPT [25] 28 July 2023	RSICap [25]	DOTA [37]	Image captioning, VQA	RSIEval [25], UCM-Captions [38], Sydney-Captions [38], RSICD [7], RSVQA-HR [35], RSVQA-LR [35]
GeoChat [26] 24 November 2023	GeoChat instruction [26]	DOTA, DIOR [39], FAIR1M [40], NWPU-RESISC45 [41], RSVQA-LR, FloodNet [42]	Scene classification, image/region captioning, VQA, visual grounding	AID [41], UCMerced [44], RSVQA-LR, RSVQA-HR, SAMRS-Bench [26], UCM-captions, RSICD
SkyEyeGPT [27] 18 January 2024	SkyEye-968k [27]	UCM-Captions, Sydney-Captions, RSICD, RSITMD [45], NWPU-Captions [46], RSVQA-HR, RSVQA-LR, RSIVQA [47], RSVG [48], DIOR-RSVG [8], DIOR, DOTA, CapERA [49], ERA [50]	Image captioning, video captioning, VQA, visual grounding	UCM-Captions, Sydney-Captions, RSICD, CapERA, RSVG, DIOR-RSVG, RSVQA-HR, RSVQA-LR
EarthGPT [28] 30 January 2024	MMRS-1M [28]	EuroSAT [51], NWPU-RESISC45, UCMerced, WHU-RS19 [52], RSSCN7 [53], FGSCR-42 [54], DSCR [55], UCM-Captions, Sydney-Captions, RSICD, RSITMD, NWPU-Captions, FloodNet, RSVQA-LR, RSIVQA, CRSVQA [56], DIOR, DOTA, FAIR1M, HRRSD [57], NWPUVHR10 [58], RSOD [59], UCAS-AOD [60], VisDrone [61], AIR-SARShip-2.0 [62], HRISD [63], SSDD [64], HIT-UAV [65], Sea-Shipping [66], Infrared-Security [67], Aerial-Mancar [68], Double-Lightvehicle [69], Oceanic Ship [70], DIOR-RSVG	Scene classification, image captioning, VQA, visual grounding, object detection	NWPU-RESISC45, CLRS [71], NASC-TG2 [72], NWPU-Captions, CRSVQA, RSVQA-HR, DIOR-RSVG, MAR20 [73]
LHRS-Bot [74] 4 February 2024	LHRS-Instruct [74], existing RS datasets, LLaVA subset [20]	NWPU-Captions, RSITMD, LHRS-Align [74], RSVQA-HR, RSVQA-LR, UCM-Captions, RSVG, DIOR-RSVG, NWPU-RESISC45, METER-ML [75], fMoW [76], RSICD	Scene classification, VQA, visual grounding	AID, WHU-RS19, NWPU-RESISC45, SIRI-WHU [77], EuroSAT, METER-ML, fMoW, RSVQA-HR, RSVQA-LR, RSVG, DIOR-RSVG, LHRS-Bench [74]
RS-LLaVA [78] 23 April 2024	RS-instructions [78]	UCM-Captions, UAV [79], RSVQA-LR, RSIVQA	Image captioning, VQA	UCM-Captions, UAV, RSVQA-LR, RSIVQA
SkySenseGPT [29] 14 June 2024	FIT-RS [29], existing RS datasets	STAR [80], NWPU-RESISC45, UCMerced, RSITMD, EarthVQA [81], FloodNet, RSVQA-LR, DOTA-v2, DIOR, FAIR1M	Image/region captioning, VQA, multilabel scene classification, relation detection/reasoning, object detection/reasoning, image/region scene graph generation, visual grounding, scene classification	FIT-RS, FIT-RSRC [29], FIT-RSFG [29], SIRI-WHU, AID, WHU-RS19, AID-multi [82], RSVQA-HR, RSVQA-LR
EarthMarker [31] 18 July 2024	RSVP [31]	NWPU-Captions, RSITMD, Sydney-Captions, DIOR-RSVG, NWPU-RESISC45, OPTIMAL 31 [83], RSD46 [59], WHU-RS19, DOTA-v2, FAIR1M, NWPUVHR10, RSOD, UCAS-AOD, VisDrone, MAR20, DOSR [84], LEVIR [85], HRSC2016 [86], HRRSD, Vaihingen [87], Potsdam [87], Hi-UCD [88], isAID [89], UAVid [90], SOTA [91], FAST [91], WHU-CD [92], DIOR	Scene classification, image/region captioning, referring object classification	UCMerced, AID, NWPU-Captions, DIOR-RSVG
VHM [30] 6 November 2024	VersaD-Instruct [30], VariousRS-instruct [30], HnstD [30]	DOTA-v2, FAIR1M, DIOR, RSVQA-LR, DIOR-RSVG, fMoW, METER-ML, NWPU-RESISC45, RSITMD, UCMerced, CrowdAI [93], BANDON [94], MtS-WH [95], MSAR [96], DeepGlobe [97]	Scene classification, VQA, visual grounding, building vectorizing, MLC, object counting, image modality, image resolution, geometric measurement	NWPU-RESISC45, METER-ML, SIRI-WHU, AID, WHU-RS19, RSVQA-LR, RSVQA-HR, DIOR-RSVG, DOTA-v2, BANDON, MtS-WH, Potsdam, MSAR, FBP [98], FAIR1M, CrowdAI, GID [99], VariousRS-instruct

(*Continued*)

TABLE 1. A COMPREHENSIVE REVIEW OF EXISTING MLLMs FOR EO ON THE INSTRUCTION-FOLLOWING AND EVALUATION DATASETS. (*CONTINUED*)

MODEL	INSTRUCTION-FOLLOWING DATASET	INSTRUCTION-FOLLOWING DATA SOURCES	EVALUATION TASKS	EVALUATION DATASETS
LHRS-Bot-Nova [100] 14 November 2024	LHRS-Instruct, LRV-Instruct [101], LHRS-Instruct-Plus [100], existing RS datasets	RSVQA-HR, RSVQA-LR, UC-Merced, fMoW, METER-ML, RSVG, DIOR-RSVG, RSICD, NWPU-Captions, RSITMD, LHRS-Align, DOTA-v2, FAIR1M, LRV-Instruct	Scene classification, VQA, visual grounding	AID, WHU-RS19, SIRI-WHU, EuroSAT, NWPU-RESISC45, METER-ML, fMoW, RSVQA-HR, RSVQA-LR, RSVG, DIOR-RSVG, LHRS-Bench
RingMoGPT [102] 04 December 2024	RingMo-VL1M [102]	AID, NWPU-RESISC45, DOTA, DIOR, HRVQA [103], LEVIR-CC [104], DOTA-GC [102], DIOR-GC [102], DOTA-Cap [102], DIOR-Cap [102], UCM-Captions, Sydney-Captions, RSICD	Image captioning, scene classification, VQA, grounded captioning, object detection, change captioning	DOTA-Cap, DIOR-Cap, UCM-Captions, Sydney-Captions, RSICD, NWPU-Captions, AID, NWPU-RESISC45, UCMerced, WHU-RS19, HRVQA, DOTA-GC, DIOR-GC, NWPU-GC [102], DIOR, DOTA, NWPUVHR10, LEVIR-CC
RSUniVLM [32] 7 December 2024	RSUniVLM-Instruct [32]	GeoChat instruction, VRSBench [36], FIT-RS, RSteller [105], RSVQA-HR, RSVQA-LR, RSITMD, RSICD, NWPU-RESISC45, UC-Merced, fMoW, DIOR-RSVG, OPT-RSVG [106], LEVIR-MCI [107], Potsdam, UAVid, LoveDA [108], OpenEarthMap [109], landcover [110], ShareGPT4V [111], AllSeeingV2 [112], DVQA [113], ChartQA [114], AI2D [115], DocVQA [116], COCO stuff [117]	VQA, scene classification, visual grounding, change captioning, change detection, semantic segmentation	RSVQA-HR, RSVQA-LR, AID, WHU-RS19, NWPU-RESISC45, SIRI-WHU, DIOR-RSVG, VRSBench, LEVIR-MCI, WHU-CD, Vaihingen, UDD5 [118], VDD [119]

MLC: multilabel classification.

the challenges posed by dataset overlap between training and testing. Our work shifts focus from architectural innovations to the fundamental issue of fair model assessment.

- We introduce ChatEarthBench, a benchmark suite containing 112K image-text pairs from previously unused data sources, enabling real zero-shot evaluation of EO MLLMs. The benchmark spans diverse tasks, including CIC/DIC, VQA, and MLC, across different spatial resolutions.
- We conduct a systematic comparative evaluation of 11 MLLMs (both general and EO-specific MLLMs) using ChatEarthBench, providing insights into their relative capabilities and limitations in EO tasks.
- Based on our benchmark results, we identify key directions for advancing EO MLLMs, including expanding EO data modalities, improving model scalability and generalizability, exploring multimodal reasoning, and enhancing reliability and robustness.

RELATED WORK

We review existing MLLMs for EO from three aspects, including model architectures, instruction-following datasets, and existing benchmarks for EO MLLMs.

MODEL ARCHITECTURES FOR EARTH OBSERVATION MULTIMODAL LARGE LANGUAGE MODELS

Most MLLMs for EO follow a standard three-component framework: a vision encoder, typically a frozen CLIP ViT [9], [122] or EVA-CLIP [123] backbone; a vision-language connector, often a linear projector or multilayer perceptron; and a pretrained LLM decoder, such as Vicuna [124] or LLaMA-2 [16], fine-tuned for EO tasks. This structure is widely adopted, with variations primarily in encoders, alignment strategies, and fine-tuning methods.

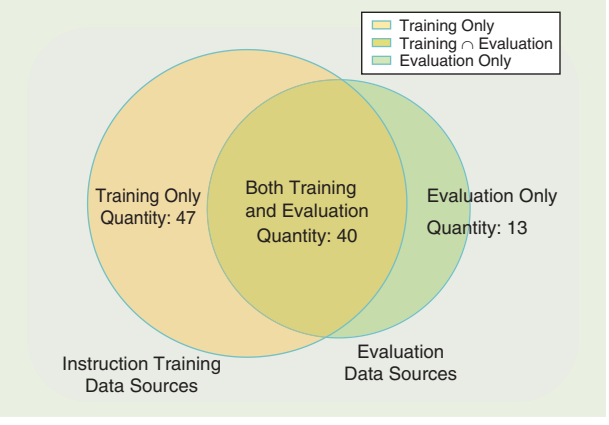


FIGURE 1. A Venn diagram of instruction training and evaluation data sources for existing representative EO MLLMs. Numbers denote dataset counts in this survey (nonexhaustive).

Several EO MLLMs follow this design with specific adaptations. RS-GPT4V [125] fine-tunes LLaVA-1.5 [126] for remote sensing applications. RS-LLaVA [78] maintains a frozen CLIP-ViT-L and Vicuna structure, employing low-rank adaptation

(LoRA)-fine-tuned alignment layers. LHRS-Bot [74] integrates learnable queries with a vision perceiver to extract visual features, which are then concatenated with language embeddings for interpretation by the LLM, enhancing multimodal understanding. EarthGPT [28] incorporates two visual encoders: DINOv2 ViT-L/14 for global features and CLIP-ConvNeXt-L for local features. It selectively unfreezes LLaMA-2 layers to improve vision-language alignment and visual grounding. EarthMarker [31] employs a mixture of visual experts to encode region-, point-, and image-level features for finer-grained spatial reasoning with text, bounding boxes, or even points.

There are also models introducing novel architectural enhancements. VHM [30] extracts multilayer CLIP features and connects them to Vicuna-v1.5, enabling fine-grained reasoning. RSUniVLM [32], built on LLaVA, introduces a Granularity-oriented Mixture of Experts (G-MoE) module to handle the multiscale nature of remote sensing imagery. G-MoE assigns inputs to three specialized experts, focusing on pixel-level, region-level, and image-level tasks. LHRS-Bot-Nova [100], an extension of LHRS-Bot, enhances its feed-forward network-based vision perceiver with a MoE architecture to improve visual feature extraction.

INSTRUCTION-FOLLOWING DATASETS FOR EARTH OBSERVATION MULTIMODAL LARGE LANGUAGE MODELS

Most studies highlight dataset scale as a key limitation in EO, where the number of image-text pairs and large-scale web-scraped data is significantly lower than in general-purpose image-text datasets. Thus, the EO community addresses this constraint by constructing diverse, multimodal, and instruction-tuned datasets.

Existing EO MLLMs primarily construct instruction-following datasets by repurposing annotated remote sensing datasets. As summarized in Table 1, different EO MLLMs are trained using instruction-following datasets built from a variety of sources. These sources originate from more than 85 existing datasets, originally designed for image captioning, VQA, object detection, scene classification, semantic segmentation, and visual grounding, among other tasks. For instance, GeoChat [26] builds its instruction-following dataset using six datasets: DOTA [37], DIOR [39], FAIR1M [40], NWPU-RESISC45 [41], RSVQA-LR [35], and FloodNet [42]. Notably, EarthGPT [28], EarthMarker [31], and RSUniVLM [32] incorporate an extensive range of existing datasets to construct more versatile instruction-following datasets, enhancing their adaptability to diverse EO tasks.

EXISTING BENCHMARKS FOR EARTH OBSERVATION MULTIMODAL LARGE LANGUAGE MODELS

For EO MLLMs, benchmarking datasets are typically introduced alongside them, and the evaluation tasks and datasets vary greatly. RSGPT [25] is evaluated on image captioning and VQA tasks, using datasets such as RSIEval [25], UCM-Captions [38], Sydney-Captions [38], RSICD [7], RSVQA-HR [35], and RSVQA-LR [35]. Some works introduce dedicated datasets for evaluation. LHRS-Bench [74] is designed to evaluate MLLMs in remote sensing using Google Earth images and manually crafted question-answer pairs. In addition, some benchmarks are specifically developed to evaluate existing EO MLLMs. VRSBench [36] is a vision-language benchmark for remote sensing, supporting image captioning, visual grounding, and VQA tasks. It consists of 29,614 remote sensing images from DOTA-v2 and DIOR,

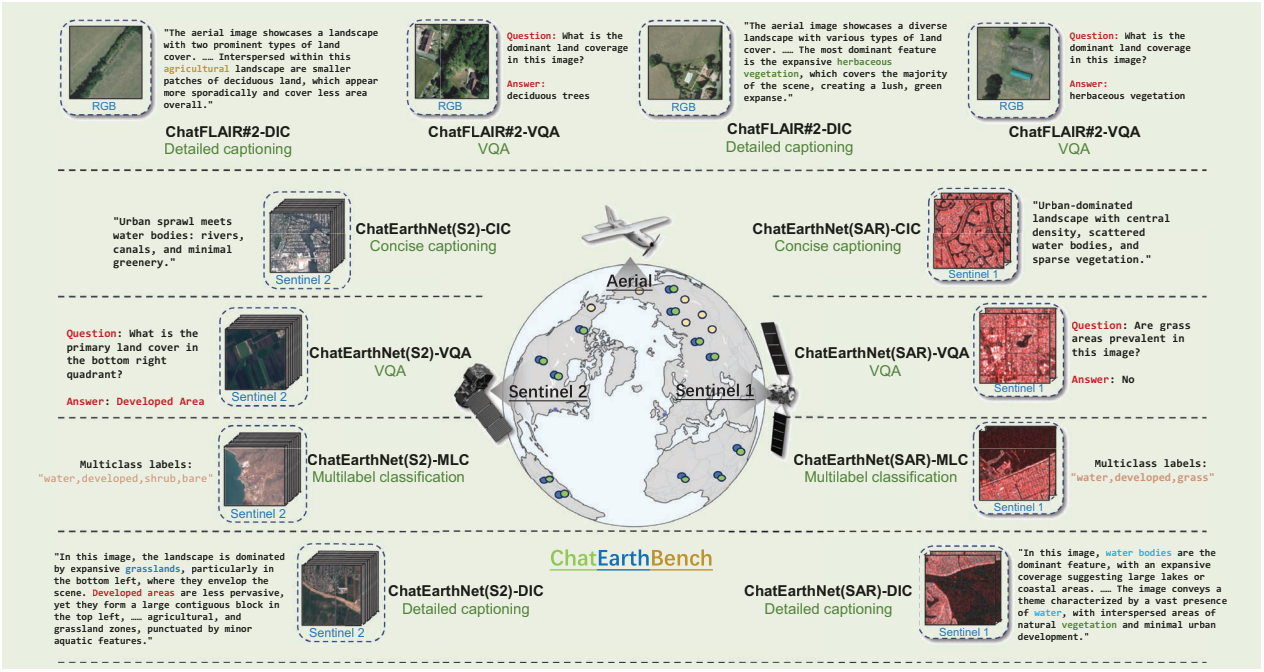


FIGURE 2. An illustration of the data samples in ChatEarthBench. There are 10 sub-datasets spanning three data modalities and four different tasks. Note that this figure is a schematic representation and does not correspond to the real geographical locations.

with human-verified captions and question-answer pairs, providing a comprehensive evaluation dataset.

Despite the availability of multiple benchmarking datasets, as shown in the Venn diagram in Figure 1, a substantial portion of the evaluation datasets overlaps with the training data of some EO MLLMs while remaining unseen by others. This inconsistency complicates fair comparisons as some models are evaluated in a zero-shot setting, while others benefit from prior exposure.

CHATEARTHBENCH FOR EVALUATING MULTIMODAL LARGE LANGUAGE MODELS

In this section, we provide a detailed overview of the construction process of ChatEarthBench and the statistical analysis of the datasets from different aspects.

CHATEARTHBENCH CONSTRUCTION

The proposed ChatEarthBench integrates data from two high-quality existing datasets: ChatEarthNet [120] and FLAIR#2 [121]. These datasets are selected due to their superior quality and absence from the training pipelines of current MLLMs. Furthermore, the two datasets complement each other in terms of spatial resolution and coverage. Specifically, ChatEarthNet emphasizes lower spatial resolution (10 m) with broader global geospatial coverage, while FLAIR#2 provides higher spatial resolution (0.2 m) and focuses predominantly on country-level regions.

The ChatEarthNet dataset [120] contains detailed long descriptions of *Sentinel-2* images. To further enrich this dataset and enable comprehensive multimodal evaluation, we integrate corresponding *Sentinel-1* images captured by the synthetic aperture radar (SAR) sensor. Specifically, we align *Sentinel-1* images with *Sentinel-2* images based on their geolocation, forming a set of 5,916 image pairs. The *Sentinel-1* images used in this dataset are in the Ground Range Detected (GRD) format and consist of two polarization channels: vertical transmit and vertical receive (VV) and vertical transmit and horizontal receive (VH), which are essential for capturing different scattering mechanisms of ground surfaces. For *Sentinel-2* images, we utilize nine spectral bands, including the red, green, and blue (RGB) bands (B2, B3, and B4) as well as bands B5, B6, B7, B8, B11, and B12. These bands capture key information across the visible (B2–B4), red-edge (B5–B7), near-infrared (B8), and shortwave infrared (B11–B12) regions of the electromagnetic spectrum. This combination of spectral bands provides rich spectral information for vegetation monitoring, land cover classification, and water or moisture assessment. In contrast, bands B1, B8A, B9, and B10 are excluded because they are primarily designed for atmospheric correction or cirrus detection, have coarser spatial resolution, or are redundant with other bands. This selection ensures a balance between spectral richness and spatial consistency.

To support the MLC task, we leverage land cover segmentation maps provided in the ESA's WorldCover project [127]. From these segmentation maps, we extract the land cover types present within each image patch as the MLC ground truth. To minimize noise from the land cover maps, land cover types accounting for less than 0.5% of the total image area are filtered out.

Regarding the text data, the original ChatEarthNet dataset offers only long detailed descriptions. In this work, we extended it by using Pixtral 12B [128] to summarize these detailed descriptions into short image captions and generate question-answer pairs for VQA tasks. Through these enhancements, we aim to provide a more diverse set of tasks for a more comprehensive evaluation. Specifically, if we denote the original description in ChatEarthNet as a *long detailed description*, the prompt for summarizing it is as follows.

Prompt for Summarizing Long Detailed Descriptions of Concise Captions

You are an artificial intelligence (AI) assistant tasked with creating a concise 30-word caption that effectively summarizes the key points of the following content: [*long detailed caption*].

To support the VQA task, we utilize the Pixtral 12B model to generate question-answer pairs based on the detailed descriptions provided in the ChatEarthNet dataset. Specifically, we design a prompt to create two types of questions as follows:

1) yes/no questions, where the answers are limited to "Yes" or "No"
2) land cover type-based questions, where the answer corresponds to a specific land cover type.

The prompt is carefully crafted to ensure that the Pixtral 12B model generates question-answer pairs strictly grounded in the given context, avoiding any fabricated or irrelevant information. The exact prompt used for generating such a VQA dataset is as follows.

Prompt for Generating Question-Answer Pairs

Generate exactly three question-and-answer pairs based on the given long description of an image. The instructions are as follows:

1) Use only the provided description to create the questions and answers.
2) Ensure that the answers are one of the following: "Yes" or "No" or a land cover type (e.g., water or tree).
3) Do not make up details or ask questions that cannot be answered directly from the description.
4) The format of output should be: Question[question id]: [question]; Answer:[answer].

The question-answer examples are as follows:

1) *Question 1:* Is tree coverage significant in the image? Answer 1: "Yes."
2) *Question 2:* What is the primary land cover in the top-left quadrant? Answer 2: Developed area.
3) *Question 3:* Are grass areas prevalent in this image? Answer 3: "No."

The context description input is: [*long detailed description*].

The variable *long detailed description* denotes the original description provided in the ChatEarthNet dataset. Note that the paired *Sentinel-1* and *Sentinel-2* image data correspond to the same land cover maps. Consequently, all generated image captions, multiclass labels, and VQA data are identically shared between the *Sentinel-1* and *Sentinel-2* visual data.

The FLAIR#2 dataset was originally developed for semantic segmentation tasks. In this work, we adopt a similar methodology to that used for constructing the ChatEarthNet dataset. Specifically, we randomly select 10,000 images from the FLAIR#2 dataset and extract semantic information from the pixel-level segmentation maps. This semantic information, along with the segmentation maps as visual inputs, is used to generate long detailed descriptions with the Pixtral 12B model.

To extract semantic information, we compute the proportion of each land type present in the image and create a text-based description to provide context for the prompt. Specifically, we identify all land type labels in the segmentation map and encode each with a unique color to represent its semantic class. The prompt includes text explicitly indicating which color corresponds to which land type. Additionally, the percentage of each land type occupying the image patch is computed and passed to the Pixtral 12B model to analyze spatial distributions and generate accurate language descriptions.

Using this semantic context, we design a prompt to generate detailed image descriptions. Note that the variable names enclosed in square brackets represent placeholders for information extracted during the semantic information processing step. The exact prompt used for dataset generation is as follows.

Prompt for Generating ChatFLAIR#2-DIC Dataset

You are an AI visual assistant that can describe the scene given a segmentation map. The map uses colors to represent different land cover types. The color legend is as follows:

- [$color_i$] color represents [$label_i$], which occupies [$percent_i$] percent area.

Do not mention any colors, color coding, or technical details. Use the given class names. Only mention land cover types in the color legend.

Generate a brief and natural description of the scene by extending: "The aerial image contains [*presented_labels*] land types." Provide a concise description on their spatial distributions.

The variables $color_i$, $label_i$, and $percent_i$ represent the color name, the corresponding land cover label, and the percentage of the image patch it occupies, respectively. The variable *presented_labels* denotes all land cover types present in the image after the filtering process. Although the prompt includes the words "brief" and "concise," they refer to different aspects; one focuses on describing the land cover types, while the other addresses their spatial distribution. When combined, the generated text forms a long and detailed description. If we include a word like "detailed" in the prompt, the generated description may become excessively long and redundant in our experiments. Thus, we design the prompt to achieve a comprehensive description by combining concise elements.

After generating the detailed and comprehensive descriptions for images in the FLAIR#2 dataset, we apply a similar prompt used in the ChatEarthNet dataset to generate question-answer pairs based on these descriptions. Furthermore, to evaluate the zero-shot counting capabilities of existing MLLMs, we manually curate 700 counting-related question–answer pairs based on the FLAIR#2 dataset. Since FLAIR#2 does not provide instance-level annotations, automatic generation of counting questions and answers is not feasible. Due to the cost of manual annotation, we select a subset of 600 high-resolution and diverse images from the dataset and manually annotate questions focusing on object- and region-specific counting. Finally, this approach results in a comprehensive dataset comprising 10 sub-datasets with four distinct tasks, as summarized in Table 2. Table 3 reports the number of auto-generated samples and the proportion that is manually checked for each subset. For ChatFLAIR#2-VQA, the reported number corresponds to the 29,892 automatically generated question-answer pairs, while the additional 700 manually curated counting-related question-answer pairs are not subject to further inspection.

CHATEARTHBENCH ANALYSIS

SUBSETS IN CHATEARTHBENCH

An overview of subsets in the ChatEarthBench is shown in Table 2, which outlines image-text pair counts, data modalities, channel counts, related tasks, spatial resolutions, and geographical coverage. These subsets are grouped under two categories: ChatEarthNet and ChatFLAIR#2. ChatEarthNet includes data derived from multispectral and SAR modalities, whereas ChatFLAIR#2 focuses on the RGB modality. In the table, we use the combination of data source and task to denote each sub-dataset. MLC and VQA are the MLC and VQA tasks, respectively. ChatEarthNet has a spatial resolution of 10 m, suitable for global-scale observations, while ChatFLAIR#2 provides a higher spatial resolution at 0.2 m, focusing on France, allowing for more localized and detailed studies. This table provides an overview of the datasets in ChatEarthBench, which is designed for evaluating the generalizability of MLLMs in EO.

WORD FREQUENCIES IN CAPTIONS AND QUESTIONS

We provide word clouds for subsets of ChatEarthBench to offer a quick and intuitive visualization of word frequencies and prominent themes within each subset. For simplicity, we use the term *ChatEarthNet-CIC* to collectively refer to both ChatEarthNet(S2)-CIC and ChatEarthNet(SAR)-CIC. Although the images in these two subsets are derived from different modalities, they correspond to the same geographic

TABLE 2. AN OVERVIEW OF THE 10 SUBSETS OF ChatEarthBench

NAME	IMAGE-TEXT PAIRS	MODALITY	#CHANNELS	TASK	RESOLUTION	COVERAGE
ChatEarthNet(S2)-DIC	5,916	Multispectral	9	DIC	10 m	Sparse global
ChatEarthNet(S2)-CIC	5,916	Multispectral	9	CIC	10 m	Sparse global
ChatEarthNet(S2)-VQA	17,748	Multispectral	9	VQA	10 m	Sparse global
ChatEarthNet(S2)-MLC	5,916	Multispectral	9	MLC	10 m	Sparse global
ChatEarthNet(SAR)-DIC	5,916	SAR	2	DIC	10 m	Sparse global
ChatEarthNet(SAR)-CIC	5,916	SAR	2	CIC	10 m	Sparse global
ChatEarthNet(SAR)-VQA	17,748	SAR	2	VQA	10 m	Sparse global
ChatEarthNet(SAR)-MLC	5,916	SAR	2	MLC	10 m	Sparse global
ChatFLAIR#2-DIC	10,000	RGB	3	DIC	0.2 m	France
ChatFLAIR#2-VQA	30,592	RGB	3	VQA	0.2 m	France

locations and therefore share identical text descriptions. The same simplification method is applied to the others as well.

The word cloud depicts frequently occurring words from the dataset, with larger words indicating a higher frequency. Figure 3 shows the word frequency for concise captions in ChatEarthNet-CIC, and prominent words include "developed area," "urban sprawl," "water bodies," and "grassland," which focus on general land use and cover categories, such as urban development and natural landscapes. In Figure 4, this word cloud emphasizes detailed descriptions of the presence and distribution of land covers from ChatEarthNet-DIC, and keywords include "developed area," "landscape," "top left," and "bottom right." Compared to Figure 3, which provides high-level summarization, Figure 4 places a stronger emphasis on spatial details and precise localization of land covers within the image. As for Figure 5, it shows the frequently occurring words in questions from VQA tasks in ChatEarthNet-VQA. Prominent words such as "land cover" and "primary land" and spatial terms like "top left" highlight the focus on querying land cover types and specific spatial regions within images.

Similarly, Figures 6 and 7 illustrate the word frequency of detailed captions and questions from ChatFLAIR#2-DIC and ChatFLAIR#2-VQA, respectively. Clearly, in Figure 6, the prominent words "aerial image," "landscape," "herbaceous vegetation," and "scattered throughout" provide observations and interpretations from aerial imagery. Moreover, in Figure 7, words such as "land coverage" and "dominant land" emphasize a focus on querying land cover types; terms like "coverage significant" and "scattered throughout" reflect questions aimed at identifying specific land features and their spatial characteristics.

ANSWER AND MULTILABEL FREQUENCIES

We also provide complementary insights into the answer and multilabel frequency. Figure 8 presents the frequency distribution of answer labels in the ChatEarthNet-VQA dataset. The most frequent answers include "Yes" (8,772 occurrences), "No" (3,544 occurrences), and "Developed Area" (2,283 occurrences), with a sharp decline in frequency for other answers. This shows an imbalance in the dataset, with a few answers dominating the distribution. Figure 9 illustrates the top 25 most frequent multilabels in the ChatEarthNet-MLC dataset. The most common multilabel,

TABLE 3. THE MANUAL INSPECTION RATE OF AUTO-GENERATED SAMPLES IN DIFFERENT SUBSETS.

SUBSETS	AUTO-GENERATED SAMPLES	MANUAL INSPECTION RATE
ChatEarthNet-DIC	5,916	100%
ChatEarthNet-CIC	5,916	100%
ChatEarthNet-VQA	17,748	100%
ChatEarthNet-MLC	5,916	30%
ChatFLAIR#2-DIC	10,000	30%
ChatFLAIR#2-VQA	29,892	30%



FIGURE 3. A word cloud for concise captions in ChatEarthNet-CIC.

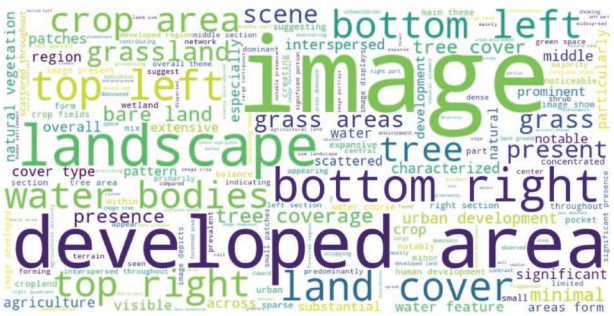


FIGURE 4. A word cloud for detailed captions in ChatEarthNet-DIC.



FIGURE 5. Word cloud for questions in ChatEarthNet-VQA.



FIGURE 6. A word cloud for detailed captions in ChatFLAIR#2-DIC.



FIGURE 7. A word cloud for questions in ChatFLAIR#2-VQA.

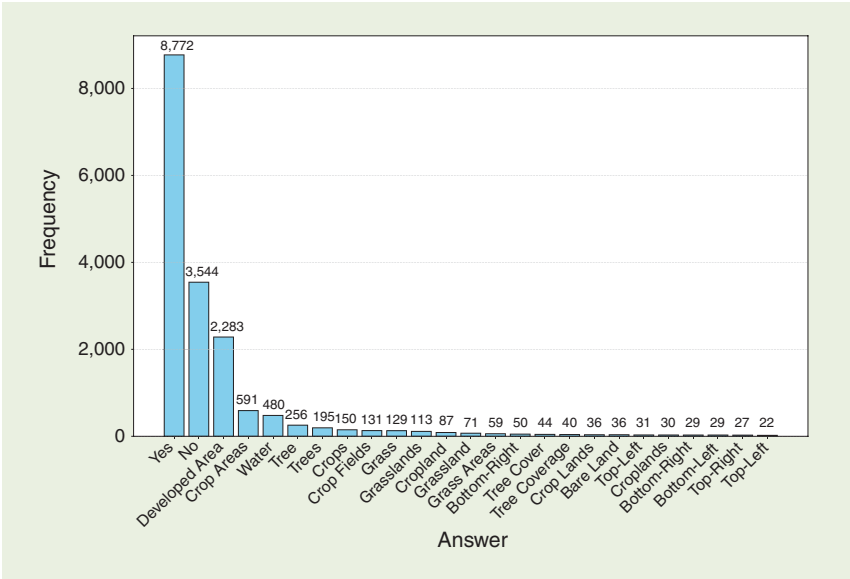


FIGURE 8. An illustration of answer frequencies in ChatEarthNet-VQA.

"water, developed, tree, grass, crop," appears 2,144 times. The distribution shows a more gradual decline, indicating relatively better diversity in multilabels compared to the answer labels.

Figure 10 shows the answer frequencies in the ChatFLAIR#2-VQA dataset, and similar to Figure 8, the most frequent answers are "Yes" (14,279 occurrences) and "No" (5,653 occurrences), which dominate the distribution significantly. Then, a decline in frequency is observed after the top two labels, with most other answers appearing fewer than 500 times.

Recent studies have highlighted the issue of language biases in VQA tasks for remote sensing. Yuan et al. [129] propose adversarial training to reduce over-reliance on frequent language priors, while Chappuis et al. [130] offer better evaluation metrics that take bias into account for both models and datasets. Motivated by these insights, we analyze our dataset distributions to better understand potential sources of bias. As shown in Figures 8 and 10, binary responses such as "Yes" and "No" are overwhelmingly dominant. Similarly, Figure 9 further shows that the distribution of multilabel combinations is heavily biased toward a few dominant patterns.

However, it is important to emphasize that all evaluations in our study are conducted in a zero-shot setting, without any fine-tuning or adaptation on the benchmark data. This evaluation protocol ensures that the frozen MLLMs cannot exploit answer frequency from the benchmark. Instead, performance reflects the models' pretraining biases and their ability to generalize to new unseen distributions. In this context, the dataset's long-tailed distribution does not introduce new biases to the models, but they can amplify preexisting language biases inherited from pretraining. In our experiments, however, we do not observe any model exhibiting systematic bias toward frequent answers. This suggests that our dataset provides a valuable benchmark for assessing the robustness of general-purpose MLLMs across both dominant and underrepresented cases.

CAPTION LENGTH

Besides word frequency, caption length is also an important aspect of analyzing a dataset by reflecting the amount of information conveyed. Figures 11–13 illustrate the caption lengths from ChatEarthNet-CIC, ChatEarthNet-DIC, and ChatFLAIR#2-DIC, respectively. In particular, in Figure 11, the word count is concentrated between 10 and 20 words, with the highest frequency around 3,500 captions having approximately 10–15 words. There are very few captions that exceed 30

words, which indicates that most captions are concise. The distribution in Figure 12 is broader, with word counts ranging from about 20 to 170 words because detailed captions are more descriptive. The highest frequency is around 70–80 words, and the distribution shows a gradual decline toward higher word counts. As for the detailed caption lengths in ChatFLAIR#2-DIC, shown in Figure 13, the word lengths are more distributed around 80–100 words, with a peak at about 90 words. Through these three figures, we can have a general overview of the caption length of these datasets.

COMPARATIVE EXPERIMENTAL ANALYSIS

A variety of MLLMs have been developed for EO tasks, each featuring unique configurations. However, only a limited number of these models have publicly released their weights. Due to the significant computational costs and challenges associated with replicating these models, we focus our evaluation on the publicly available pretrained models. To comprehensively assess the performance of existing MLLMs for EO, we select a diverse set of models from both general-purpose and EO-specific domains. Specifically, we benchmark the following models on the ChatEarthBench. This includes general-purpose MLLMs such as LLaVA-v1.5 [126], MiniGPT-4 [21], Qwen-VL-Chat [22], and InternVL2-8B [23]. These models, originating primarily from the computer vision community, have demonstrated outstanding performance across a range of vision-language tasks. However, their applicability to EO remains untested. By incorporating these models into our benchmark, we aim to evaluate their zero-shot generalization capabilities on EO tasks without any additional fine-tuning.

We also select MLLMs explicitly designed for the EO domain, including GeoChat [26], RS-LLaVA [78], VHM [30], LHRS-Bot [74], LHRS-Bot-Nova [100], SkySenseGPT [29], and RSUniVLM [32]. These models are trained on EO-specific datasets and are better aligned with the tasks in ChatEarthBench. As summarized in Table 4, we highlight the key design choices of these MLLMs evaluated in this study to provide a clearer understanding of their architectural differences and adaptation strategies. Our evaluation is conducted in a zero-shot setting, which allows us to examine how effectively these models generalize to unseen EO data and whether their pretrained weights encode sufficient transferable knowledge to handle a variety of EO tasks. The results of this evaluation will provide valuable insights into the capabilities and limitations of existing MLLMs and help to identify gaps and prioritize areas for improvement in both general-purpose and EO-specific MLLMs. Note that results for RSUniVLM

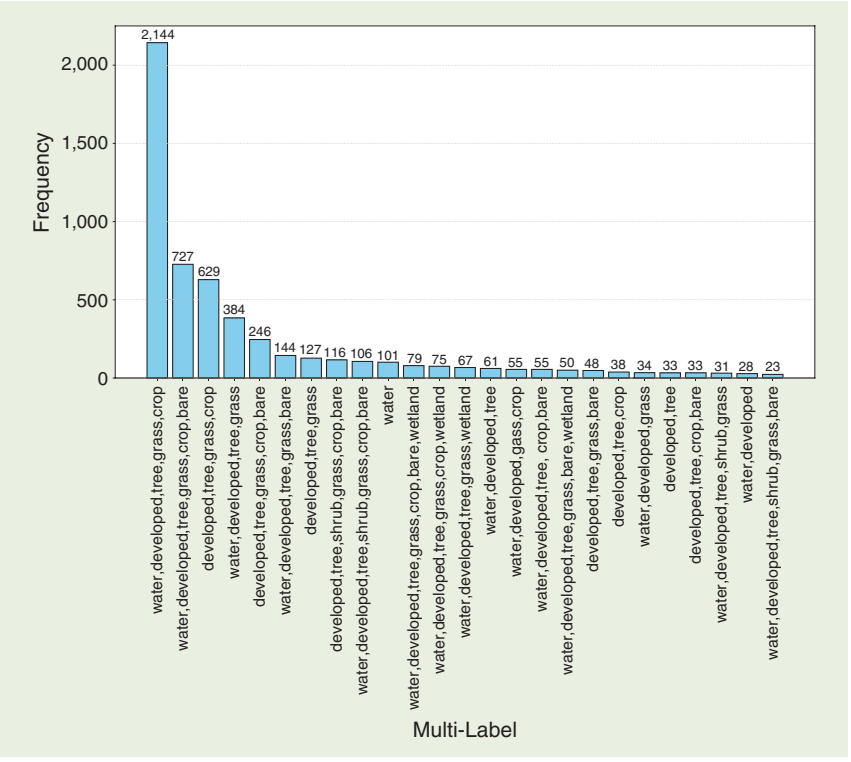


FIGURE 9. An illustration of multilabel frequencies in ChatEarthNet-MLC.

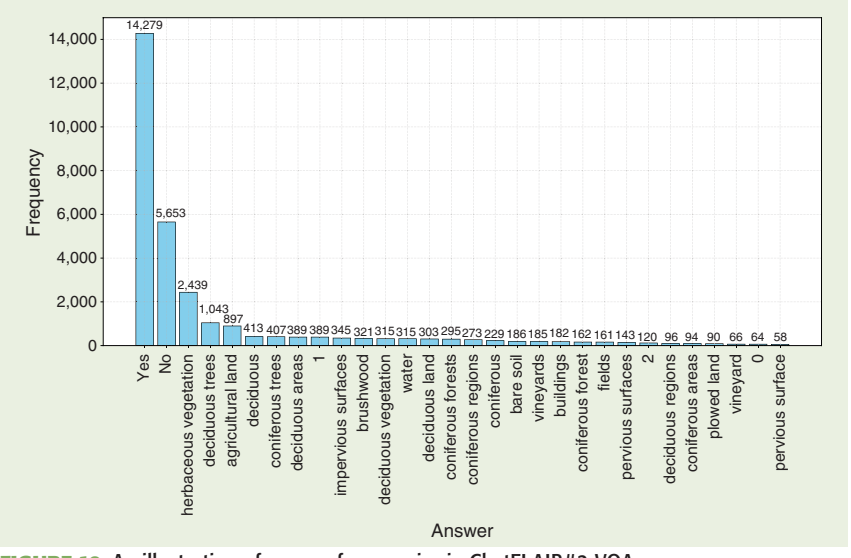


FIGURE 10. An illustration of answer frequencies in ChatFLAIR#2-VQA.

are omitted from the caption-related task tables because the model fails to reliably follow the prompt instructions.

We employ standard metrics widely used for image captioning tasks, including BLEU, CIDEr, METEOR, ROUGE-L, and SPICE, to assess model performance. It is important to note that all the existing models evaluated in this study can process only image inputs with three channels. Consequently, we use the RGB channels or replicate the last SAR channel to form three channels as input for these models. Nonetheless, our benchmark datasets are designed to remain compatible with future models that support multichannel inputs, ensuring their continued applicability for model evaluation and comparison as new architectures emerge.

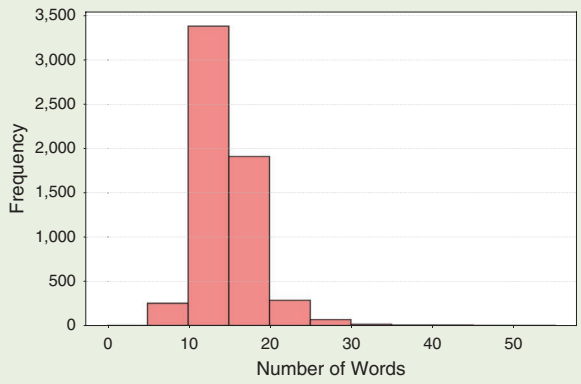


FIGURE 11. An illustration of concise caption lengths in ChatEarthNet-CIC.

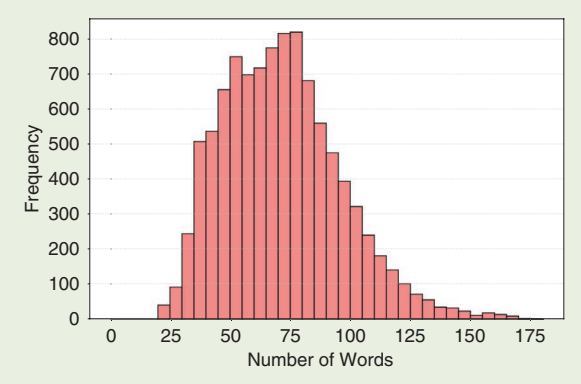


FIGURE 12. An illustration of detailed caption lengths in ChatEarthNet-DIC.

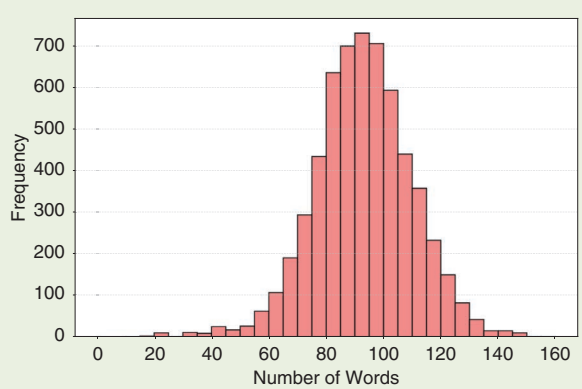


FIGURE 13. An illustration of detailed caption lengths in ChatFLAIR#2-DIC.

CONCISE IMAGE CAPTIONING TASKS

For the CIC task, we use the following prompt for all MLLMs.

Prompt #1 for the CIC Task on ChatEarthNet-CIC

Analyze the satellite image and provide a very concise description of the land covers presented in the satellite image. Possible land covers to identify are: water, developed area, tree, shrub, grass, crop, bare land, snow, wetland, mangroves, and moss. Describe only the presented ones, avoiding traversal of all.

The evaluation results for the CIC task on the ChatEarthNet (S2)-CIC dataset are summarized in Table 5. In general, the performance on image captioning tasks under the zero-shot setting is suboptimal. EO MLLMs generally outperform general MLLMs across most metrics, with RS-LLaVA achieving the highest BLEU, CIDEr, and ROUGE-L scores, while general MLLMs like Qwen-VL-Chat perform better in METEOR and SPICE. Additionally, we visualize the example outputs from the compared MLLMs in Figure 14 for a better understanding of the performance on the ChatEarthNet (S2)-CIC dataset.

When using SAR as visual input, the image captioning task becomes more challenging for the evaluated MLLMs. For this task, we use the same prompt (i.e., Prompt #1) across all models. As shown in Table 6, performance drops when switching from optical (*Sentinel-2*) to SAR data. This performance drop can be attributed to several factors. Beyond the inherent challenges of the zero-shot setting, most existing MLLMs are not adequately trained on SAR data for land cover and land use comprehension. As a result, they struggle to interpret and process SAR data. However, SAR data are crucial in EO due to their invariance to weather, illumination, and cloud conditions, making them important for many real-world EO applications. The low performance on SAR-based tasks highlights the gap between current MLLMs and practical EO requirements. For the ChatEarthNet (SAR)-CIC dataset, we also provide example outputs from the evaluated MLLMs in Figure 15. Note that the SAR image is geographically aligned with the *Sentinel-2* image shown in Figure 14. Since these MLLMs are not specifically trained on SAR data, they generally struggle to achieve satisfactory performance on this dataset.

DETAILED IMAGE CAPTIONING TASKS

For the DIC task on ChatEarthNet-DIC, we use the following prompt for all MLLMs.

Prompt #2 for the DIC Task on ChatEarthNet-DIC

Analyze the satellite image and describe the presented land covers in detail, including their types and spatial distribution. Focus only on the land covers visible in the image, such as water, developed areas, trees, shrubs, grass, crops, bare land, snow, wetlands, mangroves, or moss. Avoid listing all possible categories—mention only the ones actually observed in the image.

The performance results on the ChatEarthNet (S2)-DIC dataset are presented in Table 7. Compared to the CIC task, the performance is notably higher. This improvement is primarily because these MLLMs tend to generate lengthy descriptions that include references to land cover types, which have higher chances of matching the dataset's ground truth. However, while these scores appear higher, the current metrics have limitations in accurately measuring the semantic match between the predicted long descriptions and ground truth descriptions. Nevertheless, these metrics still provide valuable insights into the relative performance ranking of existing MLLMs on this task.

In general, EO MLLMs outperform general MLLMs. SkySenseGPT achieves the highest BLEU-1 score, while VHM leads in BLEU-2, BLEU-3, BLEU-4, and ROUGE-L score. LLaVA-v1.5 also performs well, even surpassing GeoChat. This demonstrates that LLaVA-v1.5, trained on diverse datasets, can generalize effectively to satellite imagery. We also note that some MLLMs produce excessively long descriptions. For example, InternVL2-8B generates descriptions up to 584 words, which causes issues with SPICE, failing to produce valid outputs.

On the ChatEarthNet(SAR)-DIC dataset, shown in Table 8, some MLLMs maintain a similar performance in DIC because they generate lengthy descriptions with multiple land cover types, increasing the likelihood of matching the dataset's ground truth. For example, GeoChat, VHM, and LHRS-Bot tend to enumerate all land cover types mentioned in the prompt. Although they achieve comparable scores on some metrics to those of *Sentinel-2* data, their descriptions are similar and uninformative when using SAR data.

Since certain land cover types, such as water, buildings, and bare land, share transferable visual features between optical and SAR data, some MLLMs, such as

TABLE 4. KEY DESIGN CHOICES OF THE MULTIMODAL LLMs EVALUATED IN THIS WORK.

MODEL	VISION COMPONENT	LANGUAGE COMPONENT	# LLM PARAMS.	FINE-TUNING TYPE
LLaVA-v1.5	CLIP ViT-L/14 336 px	Vicuna-v1.5-7B	7B	Full fine-tuning
MiniGPT-4	EVA-CLIP	Vicuna-7B	7B	Fine-tune linear layer
Qwen-VL-Chat	OpenCLIP ViT-bigG	Qwen-7B	7B	Full fine-tuning
InternVL2-8B	InternViT-6B 448 px	InternLLM2-8B	8B	Full fine-tuning
GeoChat	CLIP ViT-L/14 336 px	Vicuna-v1.5-7B	7B	LoRA based
RS-LLaVA	CLIP ViT-L/14 336 px	Vicuna-v1.5-7B	7B	LoRA based
VHM	CLIP ViT-L/14 336 px	Vicuna-v1.5-7B	7B	Full fine-tuning
LHRS-Bot	CLIP ViT-L/14	Vicuna-v1.5-7B	7B	LoRA based
LHRS-Bot-Nova	SigLIP-L/14	LLaMA3-8B	8B	LoRA based
SkySenseGPT	CLIP ViT-L/14 336 px	Vicuna-v1.5-7B	7B	LoRA based
RSUniVLM	SigLIP-400m	Qwen2-0.5B	0.5B	Granularity-oriented MoE (G-MoE)

Params.: parameters.

TABLE 5. A PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE ChatEarthNet(S2)-CIC SET.

MODEL TYPE	MODELS	BLEU-1	BLEU-2	BLEU-3	BLEU-4	CIDEr	METEOR	ROUGE-L	SPICE
General MLLMs	LLaVA-v1.5 [126]	11	3.3	0.9	0.2	2.6	8	13	7
	MiniGPT-4 [21]	6	1.7	0.4	0.1	0.1	7.5	7.9	5.4
	Qwen-VL-Chat [22]	14.6	4.7	1.2	0.4	4.8	**9.1**	13.6	**7.1**
	InternVL2-8B [23]	6.4	1.9	0.5	0.1	0.9	8.2	9.4	6.6
EO MLLMs	GeoChat [26]	15.7	4.8	1.2	0	7.2	6.4	12.1	6.3
	RS-LLaVA [78]	**19**	**6.8**	**2.2**	**0.7**	**9.4**	7.4	**14.8**	6.4
	VHM [30]	13.1	2.6	0.8	0	9.3	5.7	12.7	4.4
	LHRS-Bot [74]	9	2.3	0.7	0.2	4.8	4.1	7.7	3.3
	LHRS-Bot-Nova [100]	14.9	3.2	0.8	0.2	7.3	6.8	11.9	4.8
	SkySenseGPT [29]	8	2.7	0.8	0.2	0.9	8.1	10.7	6.1

Bold values indicate the best performance in each column.

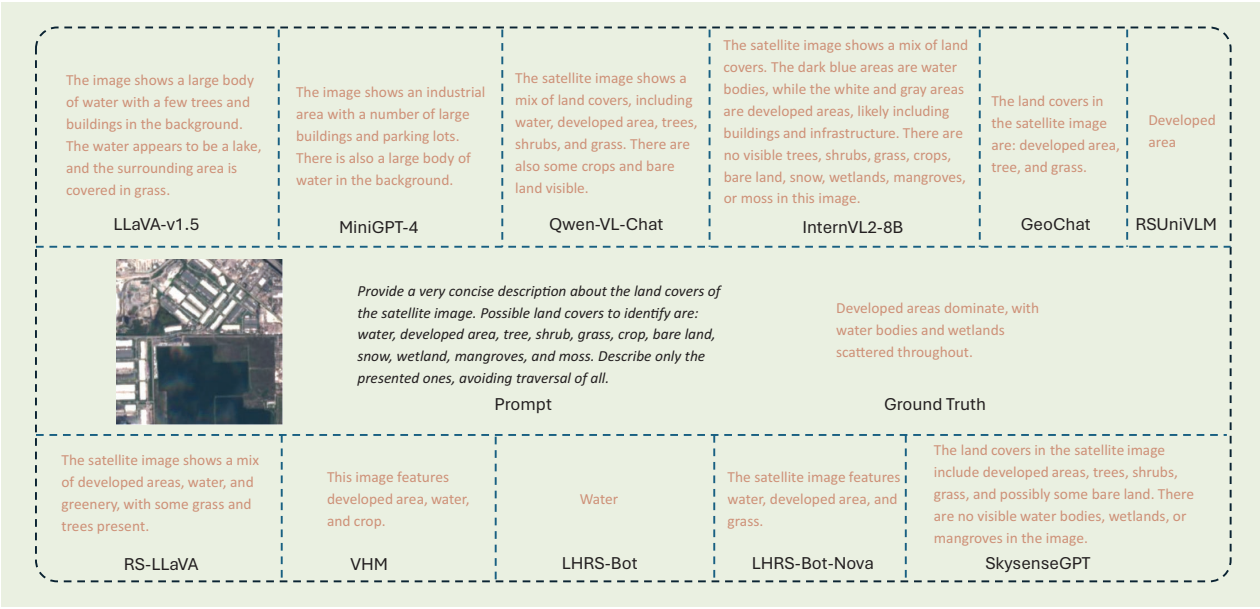


FIGURE 14. A visualization of example outputs for the concise image description task on the ChatEarthNet(S2)-CIC dataset.

TABLE 6. A PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE ChatEarthNet(SAR)-CIC SET.

MODEL TYPE	MODELS	BLEU-1	BLEU-2	BLEU-3	BLEU-4	CIDEr	METEOR	ROUGE-L	SPICE
General MLLMs	LLaVA-v1.5 [126]	4.6	1.1	0.2	0	1.3	3.8	6.4	1.9
	MiniGPT-4 [21]	5.2	1.4	0.3	0.1	0.1	6.8	6.9	4.5
	Qwen-VL-Chat [22]	12.3	3.7	0.9	0.2	1.8	**8.7**	12.3	**7.2**
	InternVL2-8B [23]	7.3	2.2	0.5	0.1	0.6	8.2	9.3	5.6
EO MLLMs	GeoChat [26]	7.9	2.5	0.6	0.1	2.3	7.5	8.8	4.8
	RS-LLaVA [78]	**17.1**	**6.4**	**2.1**	**0.6**	8.8	6.4	**13.9**	5.3
	VHM [30]	14.1	2.7	0.9	0	**10.6**	6	12.5	4.1
	LHRS-Bot [74]	11.4	2.7	0.8	0.2	6.4	4.3	9.7	1.6
	LHRS-Bot-Nova [100]	12	2.8	0.6	0.1	5	4.4	11.8	1.5
	SkySenseGPT [29]	8.3	2.8	0.7	0.2	0.4	8.3	9.4	5.4

Bold values indicate the best performance in each column.



FIGURE 15. A visualization of example outputs for the concise image description task on the ChatEarthNet(SAR)-CIC dataset.

LHRS-Bot-Nova, RS-LLaVA, and MiniGPT-4, show good generalizability and even achieve slightly better performance when transitioning from *Sentinel-2* to *Sentinel-1* data. In contrast, SkySenseGPT experiences a performance drop when transitioning from *Sentinel-2* to *Sentinel-1* imagery.

As for the DIC task on ChatFLAIR#2-DIC, the following prompt is used for all MLLMs.

Prompt #3 for the DIC Task on ChatFLAIR#2-DIC

Analyze the aerial image and describe the presented land covers in detail, including their types and spatial distribution. Focus only on the land covers visible in the image, such as [building, pervious surface, impervious surface, bare soil, water, coniferous, deciduous, brushwood, vineyard, herbaceous vegetation, agricultural land, and plowed land]. Avoid listing all possible categories—mention only the ones actually observed in the image.

The performance of the ChatFLAIR#2-DIC set is shown in Table 9. SkySenseGPT, LHRS-Bot, and LHRS-Bot-Nova perform better than other models on this dataset. Despite the improved image quality with high-resolution RGB images, the zero-shot performance of current MLLMs is still inadequate. The reason is that existing EO MLLMs are not trained with sufficient knowledge about land cover concepts. Their training predominantly focuses on object-centric understanding, such as buildings, trees, parking lots, roundabouts, and similar features.

VISUAL QUESTION ANSWERING TASKS

For the VQA task on ChatEarthNet-VQA, we apply the following prompt to all MLLMs.

Prompt #4 for the VQA Task on ChatEarthNet-VQA

[*question*]. The answer should be: [Yes] or [No] or one of the following land covers: [water, developed area, tree, shrub, grass, crop, bare land, snow, wetland, mangroves, moss].

Here, the variable *question* represents the question generated by Pixtral 12B during the dataset construction process. Compared to the image captioning task, the VQA task is relatively simpler. Considering the challenging zero-shot

TABLE 7. A PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE ChatEarthNet(S2)-DIC SET.

MODEL TYPE	MODELS	BLEU-1	BLEU-2	BLEU-3	BLEU-4	CIDEr	METEOR	ROUGE-L	SPICE
General MLLMs	LLaVA-v1.5 [126]	28.6	11.2	4	1.4	1.7	11.9	18.5	12.9
	MiniGPT-4 [21]	27.4	11.1	4	1.3	1.1	10.8	15.1	9.6
	Qwen-VL-Chat [22]	5.4	2.1	0.8	0.3	0.2	5.4	9.1	5.5
	InternVL2-8B [23]	12.9	6.4	2.8	1.2	0.1	15.6	14.2	–
EO MLLMs	GeoChat [26]	20	9.2	4.1	1.8	0	14.2	17.8	12.3
	RS-LLaVA [78]	18.2	7.4	3	1.1	0.4	10.5	16.9	13.7
	VHM [30]	30.4	**14.4**	**6.6**	**2.9**	2	13.9	**19.5**	12.9
	LHRS-Bot [74]	27.6	12.9	5.9	2.6	0.5	15.9	17.7	13.1
	LHRS-Bot-Nova [100]	30	13.3	5.9	2.5	2.1	**16.5**	18.7	**14.3**
	SkySenseGPT [29]	**33.1**	14	5.4	2	**2.9**	13.2	19.4	13.3

Bold values indicate the best performance in each column.

TABLE 8. A PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE ChatEarthNet(SAR)-DIC SET.

MODEL TYPE	MODELS	BLEU-1	BLEU-2	BLEU-3	BLEU-4	CIDEr	METEOR	ROUGE-L	SPICE
General MLLMs	LLaVA-v1.5 [126]	27.9	11	3.9	1.3	1	10.7	16.9	9.1
	MiniGPT-4 [21]	28.4	11.5	4.1	1.4	1.1	11.4	15.6	9.7
	Qwen-VL-Chat [22]	0.1	0	0	0	0	1.4	3.7	0.8
	InternVL2-8B [23]	10.9	5.3	2.2	0.8	0	14.7	12.8	–
EO MLLMs	GeoChat [26]	20	8.2	3.4	1.5	0	14.8	17.8	10
	RS-LLaVA [78]	23.9	9.5	3.8	1.4	0.9	11.8	17.9	14
	VHM [30]	28.9	12.1	5.6	2.4	**2.5**	12.2	17	13.2
	LHRS-Bot [74]	27.6	12.1	5.1	2	0.2	**16.7**	17.9	12.4
	LHRS-Bot-Nova [100]	**32.5**	**14.8**	**7.2**	**3.3**	**2.5**	16	**19.1**	**14.5**
	SkySenseGPT [29]	28	11	3.9	1.3	1.4	10.7	18.4	10.5

Bold values indicate the best performance in each column.

setting, we limit the task to two types of questions: "Yes/No" questions and questions regarding specific land cover types.

Similarly, for the VQA task on ChatFLAIR#2-VQA (excluding counting VQA), we use the following prompt for all these MLLMs.

Prompt #5 for the VQA Task on ChatFLAIR#2-VQA

[*question*]. The answer should be: [Yes] or [No] or one of the following land covers: [building, pervious surface, impervious surface, bare soil, water, coniferous, deciduous, brushwood, vineyard, herbaceous vegetation, agricultural land, plowed land].

We report three metrics for the VQA tasks: "Yes/No" accuracy, "land cover" accuracy, and "overall" accuracy, calculated as the proportion of correct answers per question type. Given the zero-shot setting and unseen datasets, we address the following specific issues:

1) For "Yes/No" questions, we truncate MLLM outputs to the first word to ensure consistency.
2) For "land cover" questions, we check if the correct answer was present, ignoring additional explanations.

The comparison results are presented in Table 10 (excluding counting VQA). Regarding "Yes/No" questions, VHM and RSUniVLM often fail to follow instructions and generate irrelevant outputs. For "land cover" questions, VHM usually outperforms others as it frequently outputs land cover types instead of defaulting to "Yes" or "No." Performance is generally better with *Sentinel-2* data compared to *Sentinel-1* (SAR), reflecting the bias of MLLMs toward RGB data. To provide a clearer understanding of model behavior, we visualize example outputs from the ChatFLAIR#2-VQA dataset in Figure 16. As shown, LLaVA-v1.5 and GeoChat fail to interpret the question correctly and do not follow the instructions, producing the answer "Yes." While several other MLLMs can identify the dominant vegetation in the image (i.e., grass), they still do not adhere to the prompt's instruction to select an appropriate label (i.e., herbaceous vegetation) from the provided candidates. In contrast, RS-LLaVA, VHM, LHRS-Bot-Nova, and SkySenseGPT successfully generate the correct instruction-compliant answer.

TABLE 9. A PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE ChatFLAIR#2-DIC SET.

MODEL TYPE	MODELS	BLEU-1	BLEU-2	BLEU-3	BLEU-4	CIDEr	METEOR	ROUGE-L	SPICE
General MLLMs	LLaVA-v1.5 [126]	32.4	13	5.4	2.4	2.4	13.1	22	12.9
	MiniGPT-4 [21]	23.7	9.1	3.2	1.1	1	10.3	14.9	9.6
	Qwen-VL-Chat [22]	14.3	6.2	2.5	1.1	0	14.4	16.1	–
	InternVL2-8B [23]	12	6.4	3.3	1.7	0	16	15.3	–
EO MLLMs	GeoChat [26]	18.8	7.6	3	1.2	0.5	12.5	17.2	12.8
	RS-LLaVA [78]	27.6	11.9	5.2	2.4	2.2	13.6	19.3	–
	VHM [30]	27.2	11.7	4.7	2	0.9	11.7	18.4	11.5
	LHRS-Bot [74]	26	13.7	**8.4**	**5.6**	1.1	16	**23.1**	16.4
	LHRS-Bot-Nova [100]	26.1	13.5	7.1	4.1	1.3	**16.2**	22.5	**16.6**
	SkySenseGPT [29]	**33.2**	**14**	5.7	2.6	**2.5**	13.5	22.5	14.9

Bold values indicate the best performance in each column.

TABLE 10. A PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE VQA TASK.

MODEL TYPE	MODELS	ChatEarthNet(S2)			ChatEarthNet(SAR)			ChatFLAIR#2 (W/O COUNTING)		
		YES/NO	LAND COVER	OVERALL	YES/NO	LAND COVER	OVERALL	YES/NO	LAND COVER	OVERALL
General MLLMs	LLaVA-v1.5 [126]	62.4	0	43.3	36.2	0	25.2	**82.9**	0.01	55.3
	MiniGPT-4 [21]	37	18	31.2	34.8	13.4	28.2	51.5	11.2	38.1
	Qwen-VL-Chat [22]	65.5	16	**50.3**	54.3	1.9	38.2	66.6	2.3	45.2
	InternVL2-8B [23]	51.8	26.6	44.1	31.5	4.1	23.1	60.4	**19.6**	46.8
EO MLLMs	GeoChat [26]	67.3	0.02	46.7	57.7	0	40.1	74	0.03	49.4
	RS-LLaVA [78]	63.7	9.9	47.3	45.9	0	31.9	78.8	1.3	52.9
	VHM [30]	7.4	**36.2**	16.2	5	**42**	16.3	5.8	14.1	8.5
	LHRS-Bot [74]	15.9	10.3	14.2	5.1	7.5	5.8	4.8	16.2	8.6
	LHRS-Bot-Nova [100]	**71.6**	1.2	50.1	**70.3**	0.02	**48.8**	82.8	9.6	**58.4**
	SkySenseGPT [29]	64.2	7.2	46.8	51.7	0	35.9	74.2	4.5	51
	RSUniVLM [32]	9.9	14	11.2	0.3	22.9	7.2	6.2	13.5	8.6

Bold values indicate the best performance in each column.

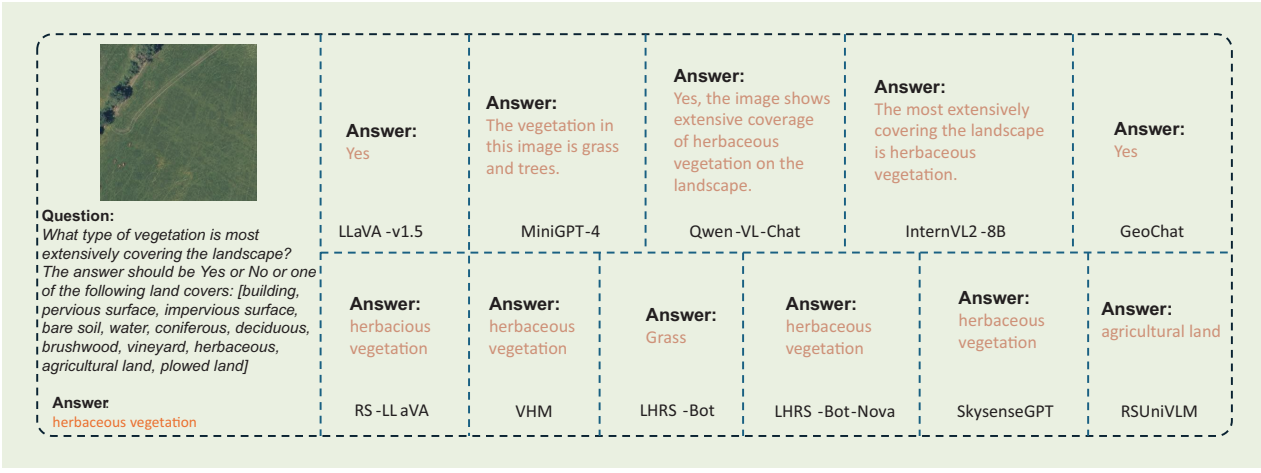


FIGURE 16. A visualization of example outputs for the VQA task on the ChatFLAIR#2-VQA dataset.

To assess the counting abilities of the evaluated MLLMs, we report their performance on the counting-related VQA subset of ChatFLAIR#2, as shown in Table 11. All MLLMs are evaluated using the same carefully designed prompt: "[question] When answering counting-related questions, respond only with the number. Do not include any other characters, words, units, or punctuation." Here, [question] is replaced by the specific counting VQA question. For evaluation metrics, "Acc." represents the proportion of predictions whose output number exactly matches the ground-truth answer, while mean absolute error (MAE) denotes the MAE between the predicted and actual counts.

Among the general-purpose MLLMs, InternVL2-8B demonstrates the strongest overall performance, achieving the highest accuracy and lowest MAE in this group. LLaVA-v1.5 achieves the second-highest accuracy. However, regarding MAE, the predictions of Qwen-VL-Chat deviate less from the ground truth, although it attains fewer exact count matches than LLaVA-v1.5. MiniGPT-4 is the weakest general model, achieving an accuracy of only 23.71%. On the EO side, LHRS-Bot, LHRS-Bot-Nova, and GeoChat perform rather poorly, showing very low accuracy and high MAE. The top general model, InternVL2-8B, surpasses all EO MLLMs in both accuracy and MAE, suggesting that large-scale general models transfer well to counting tasks on high-resolution imagery. Among EO models, RSUniVLM and VHM are the top-performing models, although they still lag behind the general MLLMs. This indicates that EO-specific instruction fine-tuning is effective for tasks such as captioning and MLC. However, incorporating more object-counting data during EO instruction fine-tuning could further strengthen numeracy capabilities and help EO models surpass the performance of state-of-the-art general MLLMs on counting-related tasks.

MULTILABEL CLASSIFICATION TASKS

For the MLC task, we use the following prompt for all MLLMs.

Prompt #6 for the MLC Task on ChatEarthNet-MLC

Classify the input into one or more of the following classes: [water, developed area, tree, shrub, grass, crop, bare land, snow, wetland, mangroves, moss]. Instructions: Choose only the relevant classes based on the input. Separate the class names with a comma (e.g., water, tree). If no class is applicable, leave the output blank. Input: [Insert input description here]. Output: [Class names separated by commas]. Only output the class names without any additional explanation.

For this task, we evaluate the performance of different MLLMs using precision, recall, and F1-score. Considering variations in the instruction-following abilities of different MLLMs, we preprocess the outputs by removing irrelevant words from the predictions before computing the metrics.

The results in Table 12 show that several MLLMs perform relatively well on the MLC task. This is largely because some

TABLE 11. A PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE COUNTING VQA TASK. ACCURACY (HIGHER IS BETTER) AND MAE (LOWER IS BETTER) ARE REPORTED.

MODEL TYPE	MODELS	ACC. (%) ↑	MAE ↓
General MLLMs	LLaVA-v1.5 [126]	65.71	0.7629
	MiniGPT-4 [21]	23.71	1.4649
	Qwen-VL-Chat [22]	59.86	0.6029
	InternVL2-8B [23]	**71.86**	**0.4943**
EO MLLMs	GeoChat [26]	33.57	1.4129
	RS-LLaVA [78]	59.29	1.3929
	VHM [30]	56	0.6886
	LHRS-Bot [74]	17.43	1.3014
	LHRS-Bot-Nova [100]	29.86	1.0543
	SkySenseGPT [29]	57.43	0.9029
	RSUniVLM [32]	61.86	0.6706

Bold values indicate the best performance in each column. Acc.: accuracy.

models can recognize specific common land cover classes, such as water, grass, trees, developed areas, and bare land. Considering the precision score, RS-LLaVA outperforms other models using both *Sentinel-2* and *Sentinel-1* data. Among general MLLMs, LLaVA-v1.5 and Qwen-VL-Chat also demonstrate good performance.

However, with SAR data, GeoChat struggles significantly, producing nearly zero results. Most of GeoChat's outputs state: "None of the classes mentioned are applicable to the input provided," indicating its inability to process SAR images in MLC tasks. In contrast, SkySenseGPT, LHRS-Bot, and RS-LLaVA achieve high performance with SAR data. Upon reviewing their predictions, SkySenseGPT demonstrates the ability to recognize classes like water, bare land, and developed areas, while RS-LLaVA identifies water, trees, grass, and bare land. These classes are relatively easier to distinguish and share visual characteristics across *Sentinel-2* and *Sentinel-1* data, suggesting that these MLLMs can generalize to SAR data to some extent.

Overall, the performance remains suboptimal in zero-shot scenarios. None of the MLLMs can reliably recognize more complex or uncommon land cover types, such as moss, snow, shrub, wetland, or mangroves. This highlights a significant gap in their knowledge of these specific land cover types and their limited ability to interpret more challenging visual inputs.

We acknowledge that ChatEarthBench is not free from dataset biases, which may arise from limited diversity in both the underlying datasets and the language annotation methods. These biases can influence evaluation outcomes, particularly regarding models' instruction-following abilities and domain coverage. However, highlighting these limitations is a key contribution of our benchmark. By carefully selecting datasets that differ from those used to train existing MLLMs for EO and evaluating models in a zero-shot setting, we aim to reveal inherent training biases and underscore the need for more balanced and comprehensive evaluation protocols. While future versions of ChatEarthBench will incorporate additional data sources to mitigate bias further, the current benchmark provides a valuable foundation for assessing the zero-shot generalization and real-world problem-solving capabilities of today's MLLMs in EO. This work also underscores the need for the community to develop more representative datasets that span diverse tasks, modalities, and geographic regions.

PERSPECTIVES ON THE FUTURE MULTIMODAL LARGE LANGUAGE MODELS FOR EARTH OBSERVATION

Based on our evaluation results, we anticipate several critical aspects that will drive the next generation of MLLMs for EO.

- *Supported data modalities in EO:* Future MLLMs should expand their ability to handle a wider range of EO data modalities. While current models can process optical imagery, next-generation models need to seamlessly integrate additional modalities such as multispectral and hyperspectral data. Incorporating these diverse inputs will allow for a more comprehensive understanding of Earth systems and improve performance on complex multimodal tasks.
- *Model scalability and generalizability:* The architectural design of future LLMs should emphasize scalability, allowing for the easy adaptation of models to new EO tasks or datasets. This includes the use of specialized domains of data that can focus on broader aspects of EO data. Researchers should also consider enhancing the in-context learning ability of the MLLMs for deploying these systems in real-world applications.
- *Multimodal reasoning:* One of the primary challenges in leveraging MLLMs for EO tasks lies in their ability to effectively understand and fuse multimodal data. For instance, enhancing existing MLLMs to integrate both *Sentinel-1* and *Sentinel-2* data and determining the optimal fusion strategy are critical for improving performance in complex and high-stakes scenarios, such as disaster response.
- *Reliability and robustness:* Reliability remains a significant challenge for MLLMs in the EO domain. Models should

TABLE 12. A PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE MULTILABEL CLASSIFICATION TASK.

		ChatEarthNet(S2)			ChatEarthNet(SAR)		
MODEL TYPE	**MODELS**	**PRECISION**	**RECALL**	**F1-SCORE**	**PRECISION**	**RECALL**	**F1-SCORE**
General MLLMs	LLaVA-v1.5 [126]	47.1	47.2	40.8	11.9	**27.9**	16.5
	MiniGPT-4 [21]	15.2	17.2	13.6	13.9	23.3	15.2
	Qwen-VL-Chat [22]	45.3	42.1	40.3	25.7	11.5	15.3
	InternVL2-8B [23]	42.2	15	21.1	0.8	0.2	0.3
EO MLLMs	GeoChat [26]	55.5	28.7	36.1	0.3	0.1	0.1
	RS-LLaVA [78]	**89.2**	33.7	47.5	**89.3**	27.6	**41**
	VHM [30]	55.4	24.7	31.5	41	17.3	24.1
	LHRS-Bot [74]	57.8	55.8	**51.9**	60.6	18.1	25.3
	LHRS-Bot-Nova [100]	42.5	**62.6**	49.3	15.1	18.8	15.8
	SkySenseGPT [29]	66.7	20.9	30.6	72.8	15.6	25.1
	RSUniVLM [32]	42.4	10	15.5	48.5	10.3	16.7

Bold values indicate the best performance in each column.

become more robust, particularly in handling noisy or incomplete data, which is common in real-world remote sensing scenarios. Additionally, ensuring that models generalize well across diverse geographic regions and environmental conditions will be key to their practical deployment.

By addressing these aspects and incorporating robust evaluation techniques, the next generation of MLLMs can be better equipped to meet the complex and growing demands of Earth monitoring and analysis.

CONCLUSION

In this work, we introduce ChatEarthBench, a benchmark designed for a fair and systematic evaluation of EO MLLMs in a zero-shot setting, thereby minimizing dataset biases. ChatEarthBench is a comprehensive evaluation dataset specifically tailored to assess the generalizability of MLLMs for EO applications. It consists of 10 image-text datasets spanning three data modalities, all of which are unseen by the evaluated MLLMs. This ensures a rigorous and unbiased zero-shot evaluation across a diverse range of real-world EO tasks. Evaluation results reveal that existing MLLMs deliver unsatisfactory performance on ChatEarthBench, indicating the need for further development to improve their generalizability, data coverage, and instruction-following abilities. Based on these results, we provide critical insights into the capabilities and limitations of current MLLMs in handling complex EO applications, aiming to foster advancements in MLLMs' development within the EO community.

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